

HydraDG — Governed Memory on HydraDB

Judge submission: value proposition, architecture, economics, scoring, evidence state, and next falsification experiments.

ONE-SENTENCE PITCH

HydraDB makes long-lived context stateful; HydraDG adds content-addressed custody, perturbation/recovery experiments, model/agent lineage, fail-closed claim governance, and reproducible economics around that context.

1. The problem we are solving

- Similarity can retrieve a semantically close fact that is stale, superseded, contradictory, or from the wrong session.
- Destructive updates hide how a state changed and make it difficult to diagnose whether an error came from retrieval, data mutation, a model, or a downstream agent.
- Long-running agents repeatedly ingest and reason over overlapping context, creating duplicated identities, context tokens, model calls, and hard-to-audit cost.
- Most benchmark summaries collapse success/failure into one score and discard inconvenient null, negative, timeout, blocked, or abstaining outcomes.

HYDRADB CONTEXT

HydraDB already attacks core memory failures with versioned temporal graphs, hybrid retrieval, entity resolution, multi-session context, and tiered storage. HydraDG is designed as a governance/evaluation layer that makes those state transitions and their downstream consequences auditable.

2. Pain point matrix: HydraDB strength + HydraDG extension

Operational pain	HydraDB strength	HydraDG extension
State confusion / stale fact served as current	Versioned temporal graph; valid-time awareness	Explicit CURRENT, CONTRADICTS, SUPERSEDED_BY, RESTORES edges; first-divergence localization.
Similarity is not relevance	Dense + sparse + graph + temporal retrieval	Candidate evidence carries custody state, source identity, claim ceilings, abstention and provenance.
Cross-session / long-horizon reasoning	Persistent structured context across sessions	Session/model/agent touches become FCG participants; error inheritance and recovery can be scored.
Duplicate memory records / repeated context	Entity resolution; tiered storage	Canonical content identity reused across contexts; deterministic occurrence/identity accounting.
Expensive AI operations on stale or bad context	Context layer can reduce dependence on raw model scale	Future avoided-call gate: stop downstream agents when a known-invalid or unresolved state should not propagate.
Unauditable benchmark or release claims	Reproducible benchmark surface	FCO lineage, SHA-256 identities, evidence classes, claim ceilings, null/negative/failure retention.

3. Golden Path architecture

8-STEP JUDGE FLOW

01 Reference → 02 Poison → 03 Antidote → 04 Hosted HydraDB → 05 Historical Results → 06 Evidence/FCG → 07 Future Work → 08 Claim Boundary

01 Reference — Declare the current valid state, source/session provenance, and content identity.

02 Poison — Inject a controlled conflicting state. Preserve predecessor identity and record the first divergent edge rather than overwriting history.

03 Antidote — Restore the valid state while keeping the poison and contradiction traversable.

04 HydraDB — Show live hosted connectivity and request-level canary traceability; do not conflate connectivity with full canonical parity.

05 Results — Show the historical full500 null/negative retrieval result and its claim ceiling.

06 Evidence — Traverse source → atom → seed → state → claim/release; show hashes establish identity, not correctness.

07 Future Work — Show BEAM 1M architecture and multi-agent/economics hypotheses as preregistered, not claimed results.

08 Claim Boundary — End on what is executed, deterministic, theoretical, future, blocked, or not established.

4. Novel state and atom scoring stack

DESIGN PRINCIPLE

Do not force every lane into one scalar. Use the scoring family only where its inputs are declared; otherwise return N/A, abstain, or preserve the failure state.

Method	Purpose	Evidence state	Current value / boundary
G^*	Synthetic state score for a frozen declared distribution	T0/T1/T2 only	T0 -0.061230; T1 0.572956; T2 -0.027496
ΔG^*	Change relative to the prior state	T0/T1/T2 only	T1 +0.634186; T2 -0.600452
Cloud Drift	$100 \times$ base-2 Jensen-Shannon divergence from frozen reference	T0/T1/T2 only	T0 0; T1 40.3629; T2 1.8729
Context-vs-Entropy	Second-stage classification of high-entropy secret-scan findings into contextual evidence classes	Executed classification lane	18,555 / 18,567 classified; 12 abstentions; 99.9354% coverage
Atom identity reuse	Exact accounting of repeated word/sentence keys	Deterministic retained accounting	65.730975% identity reuse
Anticube governance	Self/non-self $\times$ safe/unsafe classification of atoms/decisions	Future/preregistered governance signal	Not assumed to improve ranking; must be tested separately
FirstDivergenceAccuracy	Did the graph localize the earliest bad state/decision?	Future multi-agent perturbation metric	Preregistered
RecoveryRate	Did a poisoned chain return to the valid state?	Future multi-agent perturbation metric	Preregistered

5. Current evidence state

Evidence lane	State	What we can say	Claim ceiling
Historical LongMemEval full500 K=5	EXECUTED	500 total; 470 retrieval-scored; 30 abstentions. Route A Hit@5 0.9638297872 / Recall@5 0.9065957447; B/C/D lower.	Retrieval ablation only; not end-to-end QA.
Hosted HydraDB connectivity	EXECUTED / OBSERVED	Hosted API responds; hydradg + hydradg-judge-demo bound; canary relation request returns 200 with payload.	PASS_REQUEST_LEVEL, not full canonical parity.
653 FCO / 1,692 edge hosted parity	NOT_ESTABLISHED	Target exists; full scoped readback/missing-extra/root comparison not yet complete.	Do not display PASS.
Identity reuse accounting	DETERMINISTIC	31,672,976 occurrences; 10,854,020 unique; 20,818,956 duplicate occurrences.	Units are word/sentence occurrence identities, not bytes or model tokens.
Whole-corpus byte savings	NOT_MEASURED	Need frozen {path,size_bytes,sha256} acquisition manifest.	No GB-saved claim.
Energy savings	THEORETICAL_ONLY	7B scenario = $2.91465384 \times 10^{17}$ FLOPs; 0.809626 Wh equivalent under assumed efficiency.	Not wall-power measurement.
Expanded model matrix	NOT_PROMOTED	Earlier development receipts/scores were inconsistent; only receipt-backed real execution can be promoted.	Fail closed.
BEAM 1M HydraDG numbers	NONE / FUTURE	Official source identified; official rows must be revision-pinned and hashed before execution.	No HydraDG BEAM score yet.

6. Economics: what can save money, and what still needs measurement

WHY THIS MATTERS FOR HYDRADB

HydraDB publicly uses storage-based pricing and tiered storage. Current public tiers include Surge at \$25/month with 2 GB included and \$0.50/GB storage overage, and Scale at \$399/month with 10 GB included and \$0.25/GB storage overage. Canonical identity reuse is therefore economically relevant, but HydraDG must measure actual serialized bytes, metadata/edge overhead, compression, tier residency, tokens, queries and downstream model calls before converting identity reuse into a bill-savings claim.

- Identity reuse: 31,672,976 retained occurrences → 10,854,020 unique identities. The 65.730975% reuse rate is an upper-bound identity-elimination opportunity before graph-edge, metadata, compression and storage-tier overhead; it is not a 65.73% database bill reduction.
- Storage measurement to add: raw serialized bytes, canonical-object bytes, edge/metadata overhead, compressed bytes, and hot/warm/cold tier residency.
- Context measurement to add: prompt tokens before/after canonical evidence assembly, repeated evidence avoided, and context-to-answer ratio.
- Inference measurement to add: number of downstream calls blocked because an upstream atom/decision was invalid, superseded, unsupported, or abstained.
- Time abstraction: HydraDG separates semantic time (valid-time / commit-time and T0→T1→T2 perturbation chronology) from operational time. Future measurements should capture retrieval-stage latency, model wall time, recovery time, and avoided downstream wall time.
- Latency measurement to add: custody/graph overhead versus retrieval/answer accuracy and recovery quality.
- Business KPI: Cost per Correct Governed Answer = (storage + retrieval + inference + recovery) / correct final governed answers.

7. Multi-session → multi-agent perturbation experiments

RESEARCH OPPORTUNITY

HydraDB already emphasizes agent traces, temporal versioning and multi-session memory. HydraDG turns the agents/models themselves into graph participants so failures become inspectable perturbations.

- Agent chain: Retriever → Extractor → Reasoner → Decision Agent → Answer Agent.
- Every read/transform/decision creates or references an FCO; edges record READ_BY, PRODUCED, DERIVED_FROM, CONTRADICTS, SUPERSEDED_BY, RESTORES, USED_CONTEXT.
- Controlled poisons: wrong extraction, stale state, contradiction, provenance loss, and inherited bad agent decision.
- Primary metrics: ErrorPropagationRate, FirstDivergenceAccuracy, RecoveryRate.
- Guardrails: temporal reasoning, event ordering, multi-session reasoning, current-state accuracy, historical-state retention, complete evidence-path recovery.
- Economic co-metrics: context tokens, inference calls, bytes, latency, and cost per correct governed answer.

8. What we were missing - and should emphasize to judges

Missing emphasis	Why it matters
Failure localization, not just retrieval accuracy	HydraDG should be judged on whether it can point to the first bad edge/model decision and show how the error propagated.
Recovery quality	The Antidote is not just another answer; it is a measurable restoration path that retains the poison as historical evidence.
Cost of bad context	A wrong state can trigger multiple expensive downstream calls. Preventing inherited error is an economic as well as quality metric.
Canonical identity vs byte savings	Identity reuse is strong current evidence; storage-byte savings is the next measurement and should be labeled separately.
Model/agent accountability	Record which model/version/config touched each object. A model can be an evidence-producing transformation, not an opaque black box.
Abstention as a success mode	A governed memory layer should prefer abstention/blocked state over manufacturing certainty from incomplete custody.
Latency and graph overhead	If custody improves correctness but doubles latency, judges should see that tradeoff. Add wall-clock and query-stage timing.
Permissions / tenancy lineage	For production multi-agent systems, authorization context should travel with evidence paths, not be reconstructed after retrieval.
Independent replication	A future package should enable a second machine/operator to reproduce hashes, retrieval payloads, and core result tables.

9. Judge-ready answer: Why HydraDB + why HydraDG?

30-SECOND ANSWER

HydraDB gives us the right substrate: versioned, time-aware, relationship-aware context instead of flat similarity search. HydraDG adds a custody and experimentation layer around that memory. We can deliberately poison one fact, locate the first divergence, restore it without deleting history, trace which model or agent inherited the error, and keep the evidence state honest. We also content-address repeated atoms so we can measure how much duplicate identity, context, compute and eventually storage cost can be removed. Our current 65.73% identity-reuse figure is deterministic accounting; actual byte and energy savings remain explicitly separate measurement gates.

External HydraDB reference points

- HydraDB reports 90.79% overall on LongMemEval-S and describes multi-session reasoning as an open frontier even in strong systems.
- HydraDB reports 82% overall on BEAM 1M as an external reference; HydraDG does not claim that score as its own.
- HydraDB publicly positions its architecture around versioned temporal graphs, hybrid retrieval, entity resolution, tiered storage, and storage-based pricing.

10. Highest-value work before final freeze

Priority	Work	Release gate / purpose
P0	Fix/verify live UI claim state	Connectivity may PASS; full canonical parity must remain NOT_ESTABLISHED until 653/1,692 scoped readback + root comparison passes.
P0	Freeze video URL + exact release SHA	Time-stamp the preview and preserve Git/FCO lineage.
P1	Byte manifest	Freeze path,size_bytes,sha256 for acquired corpus and compute real duplicate-byte / serialized-storage savings.
P1	Context-token + inference-call instrumentation	Measure prompts, repeated evidence, avoided calls, latency and cost per correct governed answer.
P1	Official BEAM materialization	Pin official data revision; hash actual rows; execute preregistered routes only after provenance passes.
P2	Multi-agent poison/antidote extension	Run controlled agent inheritance failures and score first divergence, propagation, recovery and cost.
P2	Independent replay	Re-run the frozen artifact chain on a second environment without changing scientific results.

Sources and provenance

- HydraDB homepage: <https://hydradb.com/>
- HydraDB benchmarks: <https://benchmarks.hydradb.com/>
- HydraDB BEAM 1M report: <https://benchmarks.hydradb.com/beam.pdf>
- HydraDB “Git for Context”: <https://hydradb.com/blog/git-for-context-versioned-temporal-graphs-ai-agent-memory>
- HydraDG repository: <https://github.com/biobitworks/hydradg>
- Review branch: `hack-hydra/final-hosted-fcg-20260820`
- Current repository evidence used in this package includes SUBMISSION.md, best-use page, releaseTimepoints.ts, judge/evidence pages, and the Golden Path UI.

CRYPTOGRAPHIC BOUNDARY

This package computes SHA-256 file identities. It is NOT digitally signed and is NOT Merkle/MMR committed unless a separate real signing/commitment receipt is added.